Aakash
Medical | IIT-JEE | Foundations

Regd. Office :Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

NCERT BOOSTER TEST SERIES

Test – 10

Topics covered in various subjects:

Physics :	Atom, Nuclei, Semicondu
Chemistry :	Amines, Biomolecules,
Botany :	Complete syllabus of XI
Zoology :	Complete syllabus of XI



PHYSICS

1. In a Geiger-Marsden experiment, what is the distance of closest approach of α-particle with kinetic energy-K before it comes momentarily to rest and reverses its direction?

(1) $\frac{Ze^2}{4\pi\varepsilon_0 K}$ (2) $\frac{Ze^2}{\pi\varepsilon_0 K}$

(3) $\frac{Ze^2}{2\pi\varepsilon_0 K}$ (4) $\frac{2Ze^2}{\pi\varepsilon_0 K}$

2. It is found experimentally that 13.6 eV energy is required to separate a hydrogen atom into proton and an electron. The orbital radius of the electron in a hydrogen atom is nearly

(1) 5A° (2) 3A°

(3) 0.5A° (4) 0.6A°

3. The longest wavelength of Balmer series of the hydrogen atom is

(1) 656.3 nm (2) 1212 nm

(3) 606 nm (4) 812 nm

4. Which among the following series are in infrared region?

(1) Lyman (2) Paschen

(3) Brackett (4) Both (2) and (3)

5. Which among the following represent the Bohr radius?

(1) $\frac{h\varepsilon_0}{\pi me^2}$ (2) $\frac{h^2\varepsilon_0}{\pi me^2}$

(3) $\frac{h\varepsilon_0}{\pi me}$ (4) $\frac{h^2\varepsilon_0^2}{\pi me}$

6. A 20 kg satellite circles earth once every 4 hours in an orbit having a radius 12000 km. Assuming that Bohr's angular momentum postulate applies to satellite just as it does to an electron in the hydrogen atom, the quantum number of the orbit of the satellite will be

(1) 1.2×10^{46} (2) 1.2×10^{43}

(3) 2.4×10^{46} (4) 2.4×10^{43}

7. Using the Rydberg formula, calculate the minimum wavelength of the Lyman series.

(1) 911A° (2) 3636A°

(3) 1028A° (4) 4856A°

8. Read following statements **A** and **B** carefully and choose the correct option

Statement A: de Broglie's hypothesis that electrons have a wavelength $\lambda = h/mv$ gave an explanation for Bohr's quantised orbits by bringing in the wave- particle duality.

Statement B: Bohr's model is applicable only to hydrogenic (single electron) atoms.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Neither statement A nor B is correct

VIP HAWELI

9. The radius of innermost electron orbit of a hydrogen atom is 5.3×10^{-11} m, the ratio of radii of the $n = 3$ to $n = 4$ orbits will be

(1) 1 : 1 (2) 3 : 4

(3) 4 : 3 (4) 9 : 16

10. The ground state energy of hydrogen atom is −13.6 eV. The ratio of kinetic energy to the potential energy in the second excited state will be

(1) 1 : 1 (2) 1 : 4

(3) 1 : 2 (4) 3 : 4

11. In Bohr's model the speed of the electron in third excited state of hydrogen atom will be

(1) 5.45×10^5 m/s (2) 5.45×10^6 m/s

(3) 7.3×10^5 m/s (4) 7.3×10^6 m/s

12. The ratio of minimum frequency of Paschen series to the minimum frequency of Brackett series will be

(1) $\frac{160}{81}$ (2) $\frac{175}{81}$

(3) $\frac{175}{27}$ (4) $\frac{160}{27}$

13. A difference of 10.2 eV separates two energy levels in an atom. The frequency of radiation emitted when the atom makes a transition from the upper level to the lower level, will be

(1) 1.36×10^{14} Hz (2) 4.4×10^{14} Hz

(3) 2.46×10^{15} Hz (4) 3.2×10^{15} Hz

14. Read following statements **A** and **B** carefully and choose the **correct** option

Statement A: In Thomson's model, an atom is a spherical cloud of positive charges with electrons embedded in it.

Statement B: In Rutherford's model, most of the mass of the atom and all its positive charges are concentrated in a tiny nucleus (typically one by ten thousand the size of an atom), and the electrons revolve around it.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Neither statement A nor B is correct

15. Which among the following angular momentum is not possible for a stationary orbit?

(1) $\frac{h}{\pi}$ (2) $\frac{3h}{2\pi}$

(3) $\frac{2h}{3\pi}$ (4) $\frac{2h}{\pi}$

16. The atomic species of the same element having same chemical properties, but differ in mass are called

(1) Isomers (2) Isotones

(3) Isobars (4) Isotopes

17. James Chadwick was awarded the 1935 Nobel prize in Physics for the discovery of the

(1) Neutron (2) Electron

(3) Proton (4) Nucleus

18. The nuclides with same neutron number N but different atomic number Z, are called

(1) Isobars (2) Isotones

(3) Isotopes (4) Isomers

19. The density of the nucleus is independent of

(1) Mass number

(2) Atomic number

(3) Number of neutrons

(4) All of these

20. The energy equivalent of 5 g of substance is

(1) 9×10^{13} J (2) 18×10^{13} J

(3) 45×10^{13} J (4) 27×10^{13} J

21. The energy equivalent of four atomic mass unit in MeV is

(1) 3726 (2) 931.5

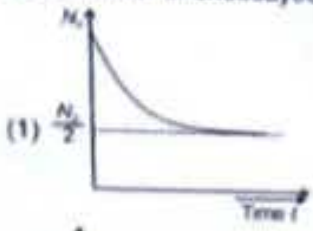
(3) 1863 (4) 1042.5

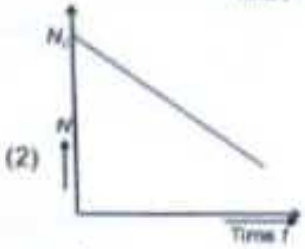
22. The binding energy per nucleon is practically independent of the atomic number for the nuclei having mass number A, where A lies as

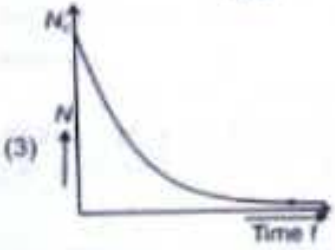
(1) $A < 30$ (2) $A > 170$

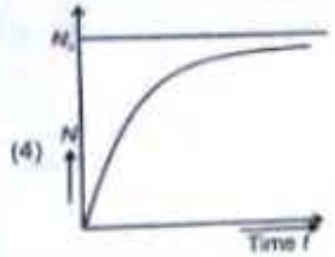
(3) $30 < A < 170$ (4) $10 < A < 30$

23. Choose the correct graph of number of undecayed nuclei versus time (where N is number of undecayed nuclei at any time t and N_0 is number of undecayed nuclei at $t = 0$)

(1) 

(2) 

(3) 

(4) 

24. The half-life of $^{238}_{92}U$ undergoing α-decay is 4.5×10^9 years. The activity of 4 g sample of $^{238}_{92}U$ will be

(1) 4.92×10^4 Bq (2) 4.92×10^3 Bq

(3) 1.23×10^7 Bq (4) 1.23×10^4 Bq

25. A radioactive sample has a half-life of 10y undergoing beta decay. The fraction of the number of nuclei undecayed after 30Y, is

(1) 0.25 (2) 0.5

(3) 0.125 (4) 0.3

26. In gamma decay which among the following quantity remains same?

(1) Mass number (2) Atomic number

(3) Number of electron (4) All of these

27. Read following statements **A** and **B** carefully and choose the **correct** option

Statement A: The basic nuclear process underlying β^- decay is the conversion of proton to neutron

Statement B: The nuclear process underlying β^+ decay is the conversion of neutron to proton

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Neither statement A nor B is correct

28. The temperature, at which two protons in a proton gas would(averagely) have enough energy to overcome the coulomb barrier, is nearly

(1) 3×10^9 K (2) 2×10^3 K

(3) 6×10^4 K (4) 4×10^5 K

29. The moderators commonly used in nuclear reactor to slow down the fast neutrons, is

(1) Heavy water

(2) Graphite

(3) Cadmium rod

(4) Both (1) and (2)

30. For α-decay of $^{226}_{88}R_a$, the daughter nuclei is

(1) $^{238}_{92}U$ (2) $^{222}_{86}R_n$

(3) $^{210}_{84}B$ (4) $^{97}_{42}M_o$

31. The range of conductivity of the semiconductor is

(1) $10^2 - 10^6$ S m^{-1}

(2) $10^2 - 10^{-4}$ S m^{-1}

(3) $10^5 - 10^{-6}$ S m^{-1}

(4) $10^{-11} - 10^{-15}$ S m^{-1}

32. Which of the following element is intrinsic semiconductor?

(1) Si (2) Ge

(3) C (4) Both (1) and (2)

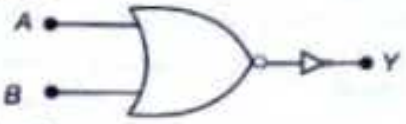
33. Which among the following is not a pentavalent impurity?

(1) Arsenic (2) Indium

(3) Boron (4) Both (2) and (3)

VIP HAWELI

44. The Boolean expression for the given logic gate will be



(1) $A \cdot B$

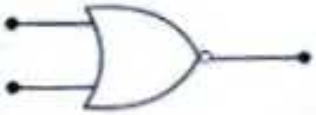
(2) $\bar{A} + \bar{B}$

(3) $A + B$

(4) $\bar{A} + B$

45. The symbolic representation of four logic gates are given below

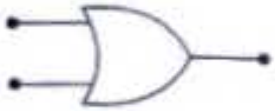
(a)



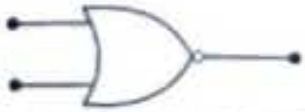
(b)



(c)



(d)



The logic symbols for OR, NOT and NAND gates are respectively

(1) (c), (b) and (a)

(2) (b), (a) and (d)

(3) (c), (a) and (d)

(4) (b), (c) and (a)

CHEMISTRY

46. Glucose on heating with HI gives

(1) n-Hexane (2) Isohexane

(3) Neohexane (4) Cyclohexane

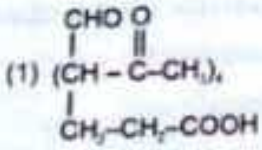
47. Number of chiral carbon in saccharic acid is

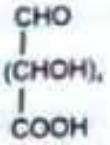
(1) 5 (2) 4

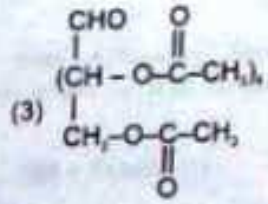
(3) 6 (4) 3

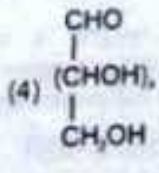
48. 

The product of the above reaction would be

(1) 

(2) 

(3) 

(4) 

49. Which one of the following statements is not true regarding glucose?

(1) The melting point of α-Form of glucose is less than the melting of β-Form of glucose

(2) Glucose forms a six-membered ring in which –OH at C-5 is involved in ring formation

(3) Glucose forms a six-membered ring in which –OH at C-6 is involved in ring formation

(4) The two cyclic hemiacetal forms of glucose differ only in the configuration of the hydroxyl group at C1, called anomeric carbon

50. The hydrolysis products of lactose are

(1) β-D-Galactose + β-D-Fructose

(2) β-D-Galactose + α-D-Glucose

(3) β-D-Glucose + β-D-Glucose

(4) β-D-Galactose + β-D-Glucose

51. Which one is animal starch?

(1) Cellulose (2) Starch

(3) Glycogen (4) Glucose

52. Fibrous proteins are
(1) Insulin and albumins
(2) Keratin and insulin
(3) Keratin and myosin
(4) Myosin and insulin

53. Vitamin B_6 is chemically known as
(1) Pyridoxine (2) Riboflavin
(3) Thiamine (4) Ascorbic acid

54. Deficiency of vitamin B_{12} causes the disease
(1) Beri beri (2) Convulsions
(3) Pernicious anaemia (4) Cheilosis

55. The enzyme that catalyses hydrolysis of maltose into glucose is
(1) Amylase (2) Maltase
(3) Zymase (4) Diastase

56. Which of the following is water soluble vitamin?
(1) Vitamin D (2) Vitamin E
(3) Vitamin C (4) Vitamin A

57. Nylon 6, 6 is an example of
(1) Polyester (2) Polyamide
(3) Polypropylene (4) Polyurethane

58. Which of the following is a step-growth polymer?
(1) Terylene (2) Orlon
(3) Teflon (4) Buna-S

59. The monomer of the polymer $\left[CH_2-\overset{Cl}{\overset{|}{C}}=CH-CH_2\right]_n$ is
(1) $CH_2=\underset{Cl}{\overset{|}{C}}H$ (2) $CH_2=CH-CH_2-Cl$
(3) $CH_2=\underset{Cl}{\overset{|}{C}}-CH=CH_2$ (4) $CH_2=\underset{Cl}{\overset{Cl}{\overset{|}{C}}}-\underset{Cl}{\overset{|}{C}}=CH_2$

60. Nylon 2-nylon 6 is prepared by the reaction between
(1) Glycine and amino caproic acid
(2) Glycine and alanine
(3) Phenol and amino caproic acid
(4) Glycine and phenol

61. Which one of the following is used to make unbreakable cups and laminated sheets?
(1) PVC
(2) Styrene
(3) Bakelite
(4) Urea-Formaldehyde resin

62. The correct order of increasing intermolecular forces is
(1) Buna-S < Polythene < Nylon 6, 6
(2) Polythene < Buna-S < Nylon 6, 6
(3) Nylon 6, 6 < Buna-S < Polythene
(4) Nylon 6, 6 < Polythene < Buna-S

63. Which of the following is used as crosslinking in vulcanisation of rubber for the manufacture of tyre rubber?
(1) 50% of sulphur (2) 5% of sulphur
(3) 5% of chlorine (4) 5% of oxygen

64. Natural rubber is a polymer of
(1) Tetra fluoroethene (2) 1, 3-butadiene
(3) Chloroprene (4) Isoprene

65. Cellulose acetate (rayon) is an example of
(1) Synthetic polymers
(2) Semi-synthetic polymers
(3) Natural polymers
(4) Both (1) and (2)

66. Serotonin is used as
(1) Antiseptic (2) Antipyratic
(3) Analgesic (4) Tranquilizer

67. Which one of the following is an example of bacteriostatic drug?
(1) Aminoglycosides (2) Ofloxacin
(3) Tetracycline (4) Penicillin

68. A broad spectrum antibiotics is active against
(1) Gram positive bacteria only
(2) Gram negative bacteria only
(3) Virus
(4) A wide range of gram-positive and gram-negative bacteria

69. Benzamide is treated separately with the following reagent. Which one of these would give benzenamine?
(1) HBr + KOH (2) NaOH + Br_2
(3) Br_2 + H_2O (4) KOH + H_2O

VIP HAWELI

70. Among the following the weakest base is
(1) Phenylmethanamine
(2) N-methylaniline
(3) N,N-Dimethylaniline
(4) Benzenamine

71. The correct increasing order of basic strength for the following compounds is

$C_6H_5NH_2$ (I) $C_2H_5NH_2$ (II) $(C_2H_5)_2NH$ (III) NH_3 (IV)

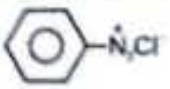
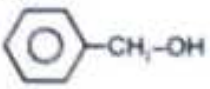
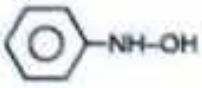
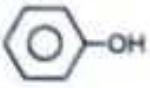
(1) (I) < (II) < (III) < (IV)
(2) (III) < (II) < (I) < (IV)
(3) (I) < (IV) < (II) < (III)
(4) (I) < (II) < (IV) < (III)

72. The major organic product (P) formed in the following reaction

$CH_3NH_2 + C_6H_5COCl \rightarrow P$, is

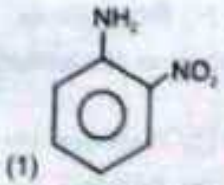
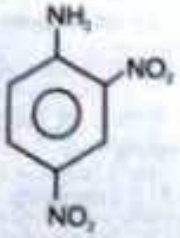
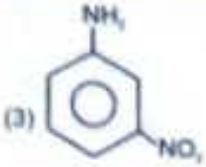
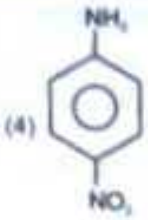
(1) $C_2H_5NHCOC_6H_5$ (2) $CH_3NHCOC_2H_5$
(3) $CH_3NHCOC_6H_5$ (4) CH_3NH_2

73. Reaction of Benzenamine with $NaNO_2$ + HCl, followed by acidic hydrolysis give

(1)  (2) 
(3)  (4) 

74. Which of the following is soluble in alkali?
(1) N,N-Diethylbenzenesulphonamide
(2) N–Ethylbenzenesulphonamide
(3) N,N–Dimethylbenzenesulphonamide
(4) N,N-Diphenylbenzenesulpheramicle

75. During the reaction of aniline with conc. HNO_3 and conc. H_2SO_4, the major product formed is

(1) 
(2) 
(3) 
(4) 

76. Which of the following amines will give most easily N_2 gas on treatment with $NaNO_2$ + HCl at 275 K?

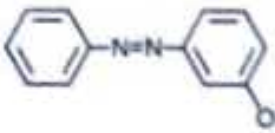
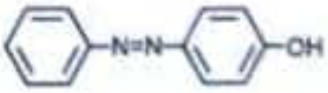
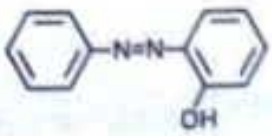
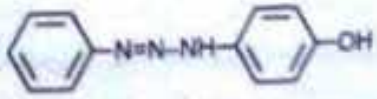
(1) $C_2H_5NH_2$ (2) $C_6H_5NHCH_3$
(3) $C_6H_5NH_2$ (4) $(CH_3)_3N$

77. Aromatic fluoride can be obtained by which of the following reaction?

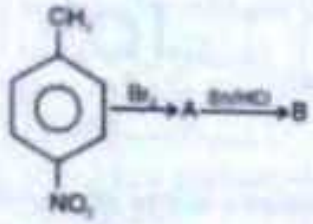
(1) ArN_2Cl +
(2) $ArN_2Cl + HF \longrightarrow$
(3) ArN_2Cl + 
(4) $ArNH_2 + HF \longrightarrow$

78. In the following reaction, the product (P) is

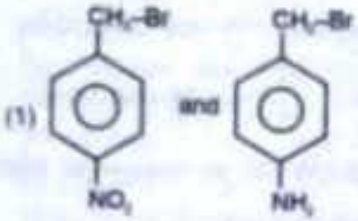


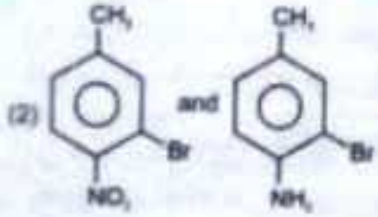
(1) 
(2) 
(3) 
(4) 

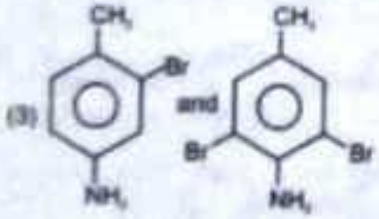
79. In the reaction

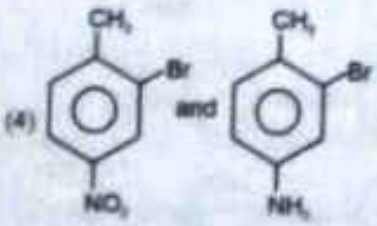


A and B is

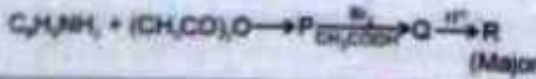
(1) 

(2) 

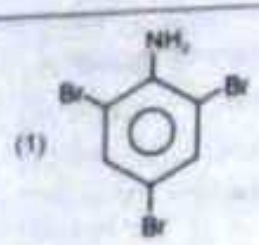
(3) 

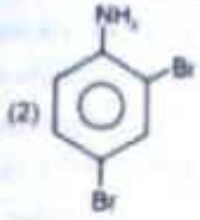
(4) 

80. The final product R obtained in given reaction sequence is

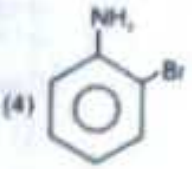


(Major)

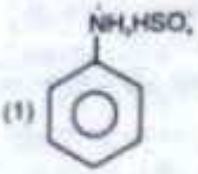
(1) 

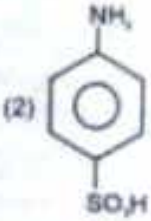
(2) 

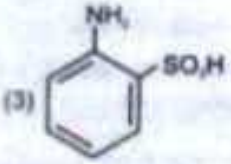
(3) 

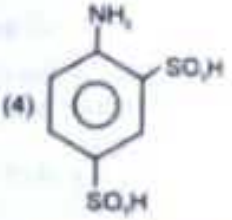
(4) 

81. The product formed by the reaction of aniline with concentrated sulphuric acid is

(1) 

(2) 

(3) 

(4) 

82. When primary amine reacts with Chloroform and ethanolic KOH, the reaction is called

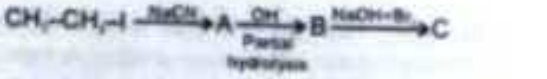
(1) Mendius reaction

(2) Hoffmann's reaction

(3) Carbylamine reaction

(4) Coupling reaction

83. In the following sequence of the reactions, the product C is



(1) Ethanamine (2) Methanamine

(3) Ethanamide (4) Propanamide

84. Which of following is used as an antifertility drug?

(1) Iproniazid (2) Norethindrone

(3) Bithionol (4) Terpineol

85. Artificial sweetner which is unstable at cooking temperature is

(1) Aspartame (2) Sucralose

(3) Sucrose (4) Alitame

86. Which one of the following is cationic detergent?

(1) Sodium dodecyl benzene sulphonate

(2) Sodium acetate

(3) Cetyltrimethyl ammonium bromide

(4) Sodium laurylsulphate

87. Which one of the following is employed as a antacids?

(1) Equanil (2) Seldane

(3) Ranitidine (4) Valium

88. Indentify the correct order of solubility in water

(1) $C_2H_5NH_2 > (C_2H_5)_2NH > C_6H_5NH_2$

(2) $C_6H_5NH_2 > (C_2H_5)_2NH > C_2H_5NH_2$

(3) $C_2H_5NH_2 > C_6H_5NH_2 > (C_2H_5)_2NH$

(4) $(C_2H_5)_2NH > C_6H_5NH_2 > C_2H_5NH_2$

89. The correct IUPAC name for the compound $(CH_3\text{-}CH_2)_2NCH_3$ is

(1) N-methyl-N-ethylethanamine

(2) N-ethyl-N-methylethanamine

(3) N,N-diethylmethanamine

(4) N-methylbutanamine

90. By which of the following reactions aniline can be prepared?

(1) Reduction of benzonitriles

(2) Hoffmann bromamide degradation reaction

(3) Gabriel phthalimide synthesis reaction

(4) Reduction of benzamide

BOTANY

91. All of the given are defining properties of living organisms, **except**

(1) Growth

(2) Metabolism

(3) Cellular organisation

(4) Consciousness

92. Match the Column-I with Column-II and select the **correct** option

Column-I	Column-II
(A) Botanical garden	(i) Provide information for identification of names of species found in an area
(B) Museum	(ii) Collection of living plants for reference
(C) Manuals	(iii) Contain actual account of habitat and distribution of plants of a given area
(D) Flora	(iv) Collection of preserved plants and animals specimens for study and reference

	(A)	(B)	(C)	(D)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iv)	(iii)	(ii)	(i)
(3)	(ii)	(iv)	(i)	(iii)
(4)	(ii)	(iv)	(iii)	(i)

93. Which of the given protozoans causes the disease sleeping sickness?

(1) *Paramoecium*

(2) *Trypanosoma*

(3) *Plasmodium*

(4) *Entamoeba*

94. The parasitic fungi causing white rust in mustard is

(1) *Albugo* (2) *Rhizopus*

(3) *Penicillium* (4) *Colletotrichum*

95. All of the given pteridophytes belong to class Pteropsida, **except**

(1) *Pteris* (2) *Dryopteris*

(3) *Equisetum* (4) *Adiantum*

96. Which of the given structures w.r.t. an angiosperm are haploid?

(A) Microspore

(B) Embryo

(C) Microsporangium

(D) Egg cell

(1) (B) and (C) (2) (A) and (C)

(3) (C) and (D) (4) (A) and (D)

97. In monocotyledonous seeds, the plumule and radicle are enclosed in sheaths which are called A and B respectively.

Select the **correct** option which fills A and B.

(1) A- Coleoptile B- Coleorhiza

(2) A- Coleorhiza B- Coleoptile

(3) A- Scutellum B- Aleurone layer

(4) A- Aleurone layer B- Scutellum

98. Flowers of which of the given plants have bilateral symmetry?

(1) Mustard (2) Cassia

(3) Canna (4) Chilli

99. Read the statements stating them true (T) or false (F) and select the **correct** option.

(A) Phloem transport food material usually from leaves to other parts of the plants.

(B) Gymnosperms lack vessels in their xylem.

(C) Xylem parenchyma cells are living and thick-walled.

	A	B	C
(1)	F	T	T
(2)	T	F	T
(3)	F	T	F
(4)	T	T	F

100. Select the **incorrect** statement w.r.t. sapwood.

(1) It is involved in conduction of water and minerals from root to leaf

(2) It is the peripheral region of the secondary xylem

(3) It is lighter in colour

(4) It has elements with highly lignified cell walls

101. Which of the given cell organelles is important site of formation of glycoproteins and glycolipids?

(1) Golgi apparatus

(2) Endoplasmic reticulum

(3) Lysosome

(4) Vacuole

102. Select the **incorrect** match.

(1) Typical bacteria – 1 to 2 µm

(2) Mycoplasma – 0.3 µm

(3) Viruses – 0.02 to 0.2 µm

(4) A typical eukaryotic cell – 2 to 5 µm

103. Which of the following events occur(s) during S phase of cell cycle?

(A) Duplication of centriole

(B) DNA replication

(C) Protein synthesis

(1) Only (A) (2) (A) and (B)

(3) Only (C) (4) Only (B) and (C)

104. Dissolution of synaptonemal complex occurs at the beginning of which of the given stages of prophase I of meiosis?

(1) Zygotene (2) Pachytene

(3) Diplotene (4) Diakinesis

105. Active transport differs from facilitated transport as the former

(1) Requires special membrane protein

(2) Is highly selective

(3) Uses energy to pump molecules against a concentration gradient

(4) Is the case in which transport saturates

106. Evaporative loss of water by plants is known as

(1) Transpiration (2) Guttation

(3) Imbibition (4) Plasmolysis

107. Deficiency of which of the given sets of elements in plants cause inhibition of cell division?

(1) N, K, S and Mo

(2) Ca, Mg and Cu

(3) Ca, Fe and Ni

(4) Cu, Ca and Mn

108. Read the statements and select the **correct** option.

Statement A: Potassium helps to maintain an anion-cation balance in cells.

Statement B: Magnesium helps to maintain the ribosome structure.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

109. Cyclic photophosphorylation

(1) Is also known as Z-scheme

(2) Results in synthesis of ATP, $NADPH + H^+$ and O_2

(3) Is associated with splitting of water

(4) Occurs in stroma lamellae

110. In C_4 plants, photorespiration does not occur. This is because

(1) They have a mechanism that increases CO_2 concentration at the enzyme site

(2) Photorespiration is a wasteful process

(3) Photorespiration is not associated with synthesis of sugar

(4) Photorespiration does not release ATP

111. Which of the given is also known as complex IV w.r.t. electron transport system (ETS)?

(1) NADH dehydrogenase

(2) Cytochrome bc_1 complex

(3) Cytochrome *c* oxidase complex

(4) ATP synthase

112. Yeasts poison themselves to death when the concentration of alcohol reaches about

(1) 7% (2) 13%

(3) 2% (4) 29%

113. Which of the given plant hormones helps to initiate rooting in stem cuttings?

(1) Auxin (2) Gibberellic acid

(3) Ethylene (4) Abscisic acid

114. The response of plants to period of day/night is termed as

(1) Photoperiodism (2) Vernalisation

(3) Plasticity (4) Vivipary

115. Which amongst the given organisms have minimum life-span?

(1) Parrot (2) Butterfly

(3) Crocodile (4) Crow

116. Read the following statements stating them true (T) or false (F) and select the **correct** option

(A) In several simple plants like algae, bryophytes and pteridophytes, water is the medium through which gamete transfer takes place

(B) Both of the fusing gametes are motile in a few fungi and algae

(C) In seed plants, pollen grains are the carriers of male gamete

	(A)	(B)	(C)
(1)	F	F	F
(2)	F	F	T
(3)	F	T	T
(4)	T	T	T

117. All of the given are water pollinated plants, **except**

(1) *Vallisneria* (2) *Hydrilla*

(3) Water lily (4) Sea grass

118. In mature seeds of which of the given plants, endosperm persists and may be used up during seed germination?

(1) Castor (2) Pea

(3) Groundnut (4) Beans

119. The inheritance of flower colour in the dog flower is an example of

(1) Co-dominance

(2) Incomplete dominance

(3) Multiple allelism

(4) Pleiotropy

120. How many chromosomes are there in a cell of human male who is inflicted with Klinefelter's syndrome?

(1) 46 (2) 45

(3) 47 (4) 44

121. In human genome, chromosomes with most and fewest numbers of genes are respectively

(1) Chromosome X and Chromosome Y

(2) Chromosome 1 and Chromosome Y

(3) Chromosome Y and Chromosome X

(4) Chromosome Y and Chromosome 1

VIP HAWELI

122. The process of polymerisation of amino acid to form a polypeptide is called

(1) Replication

(2) Transcription

(3) Translation

(4) Reverse transcription

123. *Saccharum barberi*

(1) Is a tropical cane grown in South India

(2) Had thicker stems

(3) Had higher sugar content

(4) Had poor yield

124. Pusa Swarnim is bred by hybridisation and selection, for disease resistance to white rust is a variety of which of the given crop?

(1) Wheat (2) *Brassica*

(3) Cow pea (4) Chilli

125. Which of the given bioactive molecules is given to the patients who have undergone myocardial infarction leading to heart attack?

(1) Streptokinase (2) Cyclosporin A

(3) Lipase (4) Statin

126. Which of the given enzymes is helpful in removing oily stains from the laundry?

(1) Lipase

(2) Pectinase

(3) Protease

(4) Amylase

127. Numbers of birth during a given period in the population that are added to the initial density is called

(1) Natality (2) Mortality

(3) Immigration (4) Emigration

128. An orchid growing as an epiphyte on a mango branch, is an example of

(1) Commensalism (2) Amensalism

(3) Parasitism (4) Mutualism

129. Refer the following statements and select the **correct** option.

Statement A: Secondary succession is faster than primary succession.

Statement B: Xerarch succession takes place in dry areas.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

130. Consider the statements stating them true (T) or false (F) and select the **correct** option.

(A) Except for deep sea hydro-thermal ecosystem, sun is the only source of energy for all ecosystems on earth.

(B) All organisms are dependent for their food on producers, either directly or indirectly.

(C) In an aquatic ecosystem, GFC is the major conduit for energy flow.

(D) The detritus food chain (DFC) begins with dead organic matter.

	(A)	(B)	(C)	(D)
(1)	F	F	T	T
(2)	T	T	F	F
(3)	F	F	F	F
(4)	T	T	T	T

131. India has more than ___A___ genetically different strains of rice, and ___B___ varieties of mango.

Select the option which **correctly** fills A and B.

(1) A – 50,000 B – 1,000

(2) A – 1,000 B – 50,000

(3) A – 5000 B – 10,000

(4) A – 10,000 B – 5,000

132. Match the Column-I with Column-II w.r.t. extinct species and their native area and select the **correct** option.

Column-I	Column-II
(A) *Dodo*	(i) Australia
(B) Quagga	(ii) Mauritius
(C) Thylacine	(iii) Russia
(D) Steller's sea cow	(iv) Africa

	(A)	(B)	(C)	(D)
(1)	(ii)	(iv)	(i)	(iii)
(2)	(iii)	(i)	(ii)	(iv)
(3)	(ii)	(iii)	(i)	(iv)
(4)	(iv)	(ii)	(iii)	(i)

133. In India, the Air (Prevention and control of pollution) Act came into force in

(1) 1981 (2) 1987

(3) 1976 (4) 1971

134. Which of the given greenhouse gases has least relative contribution to total global warming?

(1) CH_4 (2) CO_2

(3) N_2O (4) CFC

135. During which phase of meiosis I, the bivalents arrange on the equatorial plate?

(1) Prophase I (2) Metaphase I

(3) Anaphase I (4) Telophase I

ZOOLOGY

136. Read the following statements and select the **correct** option

(a) The coelenterates have tentacles which bear cnidoblasts

(b) Animals which bear fins for locomotion are grouped under class Pisces

(c) Hemichordates exhibit external fertilisation and direct development

(1) (a) and (b) are correct, (c) is incorrect

(2) (b) and (c) are correct, (a) is incorrect

(3) (b) and (c) are incorrect, (a) is correct

(4) (a) and (b) are incorrect, (c) is correct

137. Which of the following is the **correct** matching of an animal with its characteristic and taxon?

	Animal	Characteristic	Taxon
1.	*Trygon*	Ventral mouth. Females bear clasper. viviparous	Chondrichthyes
2.	*Petromyzon*	Shows migration, placoid scales are present, closed type circulation	Cyclostomata
3.	*Exocoetus*	Fertilisation is external, terminal mouth, air bladder is present	Osteichthyes
4.	*Ichthyophis*	Possess external ear opening, cloaca is present, eyes are without eyelids	Amphibia

138. Choose the **mismatch** w.r.t. animal and its characterstic

(1) *Ancylostoma* – Muscular pharynx

(2) *Hirudinaria* – Circular and longitudinal muscles help in locomotion

(3) *Sepia* – Anterior head region has sensory tentacles

(4) *Pleurobrachia* – Triploblastic and acoelomate

139. Select the **incorrect** statement w.r.t. cockroach

(1) Two pairs of wings are present, one pair each on 2nd and 3rd segment of thorax

(2) A pair of salivary gland is present near crop

(3) Nervous system is represented by segmentally arranged ganglia and ventral nerve cord

(4) Forewings and hind wings are present before last moulting

140. Function of __A__ epithelium is secretion and absorption. This is commonly found in ducts of __B__. Choose the option which fill the blanks A and B **correctly**

	A	B
1.	Sqaumous	Endocrine
2.	Ciliated	Exocrine
3.	Cuboidal	Glands
4.	Compound	Pancreas

VIP HAWELI

141. Consider the following statements and choose the **correct** answer.

(A) Phospholipids contain, in addition, a phosphorylated nitrogenous compound

(B) The binding of the substrate induces the enzyme to alter its shape, fitting more tightly around the substrate

(C) Starch, cellulose and Insulin are polymer of Monosaccharides from above statements

(1) All are correct

(2) All are incorrect

(3) Only A and B are correct

(4) Only B and C are correct

142. Enzyme carbonic anhydrase facilitates the following reaction in both direction $CO_2 + H_2O \xrightarrow{\text{Carbonic anhydrase}} H_2CO_3 \xrightarrow{\text{Carbonic anhydrase}} HCO_3^- + H^+$ at the alveolar site where __A__, the reaction proceeds in the __B__.

Select the option that **correctly** fills the blanks A and B.

(1) pCO_2 is high, opposite direction

(2) pO_2 is high, forward direction

(3) pCO_2 is low, opposite direction

(4) pO_2 is low, opposite direction

143. Which of the following statement is **correct** for human dentition?

(1) All premolars are diphyodont

(2) Each jaw possess four molar teeth in adult humans

(3) In each jaw all four incisors are diphyodont

(4) 2nd molar teeth in each jaw is consider as wisdom teeth in adults

144. If a person consumes boiled sweet potato, its digestion will start in

(1) Stomach

(2) Buccal cavity

(3) Small intestine

(4) Large intestine

145. In spirometry of healthy adult, the value of FRC is always

(1) More than VC (2) Less than RV

(3) Equal to TLC (4) More than the ERV

146. The SAN generates a new action potential

(1) At the initiation of ventricular systole

(2) At the end of ventricular diastole

(3) At the end of joint diastole

(4) At the end of atrial systole

147. Under normal physiological conditions, the volume of blood pumped out by each ventricle of human heart per five minute is

(1) 40 ml (2) 5 L

(3) 60 ml (4) 25 L

148. In most of the invertebrates, the excretory structures are present in simple tubular forms, green glands perform the excretory function in

(1) *Planaria* (2) Prawns

(3) *Amphioxus* (4) Rotifers

149. All of the following are **correct** for function of PCT **except**

(1) Reabsorption of HCO_3^-

(2) Reabsorption of H^+

(3) Maximum reabsorption of water

(4) Secretion of HN_3

150. Which of the following part(s) of nephron has osmolarity less than 300 $mOsmolL^{-1}$?

(1) PCT and collecting duct

(2) PCT only

(3) Loop of Henle

(4) DCT

151. Select the **incorrect** statement w.r.t. locomotion in different organisms?

(1) *Hydra* uses its tentacles for locomotion

(2) In *paramecium*, cilia helps in locomotion

(3) In human all muscles help in movement that lead to displacement of the body

(4) Pseudopodia helps in movement of leucocytes

152. How many structures, given below in the box, are present in appendicular skelton?

Clavicle, Sternum, Coccyx, Lacrimal, Zygomatic, Tarsal Vertebrochondral rib

(1) Two (2) Four

(3) Five (4) Six

VIP HAWELI

153. Identify the **incorrect** statement w.r.t. evolution

(1) Industrial melanism is an example of evolution by anthropogenic action

(2) Antibiotic resistance in bacteria supports the evolution by natural selection

(3) Evolution by natural selection is the directed process in the sense of determinism

(4) Evolution is a stochastic process based on chance mutation in organism and chance events in nature

154. Which of the following is a palindromic sequence?

(1) 5'— GAACCC— 3'
3'— CTTGGG— 5'

(2) 5'— CTATCG— 3'
3'— GATAGC— 5'

(3) 5'— ATCGAT— 3'
3'__ TAGCTA__ 5

(4) 5'— CTTGAA— 3'
3'— GAACTT— 5'

155. Which of the following is not **true** w.r.t. cloning vector pBR322?

(1) It does not possess any restriction site for *EcoRI* in amp^R gene

(2) *Hind* III restriction site is excluded from amp^R gene and tet^R gene

(3) amp^R gene codes the antibiotic ampicillin

(4) amp^R gene and tet^R gene help in identifying and eliminating non-transformant and selectively permitting the growth of the transformant.

156. Read the following statements A and B and select the **correct** option.

Statement A: The Step elution is used for isolation of genetic material in the process of recombinant DNA technology.

Statement B: If insertional inactivation occurs in *Lac Z* gene the colonies of non-recombinant do not produce any colour.

(1) Both the statements are incorrect

(2) Both the statements are correct

(3) Only statement A is incorrect

(4) Only statement B is incorrect

157. Match the Column-I and Column-II and select the **correct** answer

Column-I		Column-II
(A) cry II Ab	(i)	Reduce post-harvest losses
(B) GM plants	(ii)	Powerful technique to identify many genetic disorder
(C) PCR	(iii)	Controls the cotton boll-worms
(D) α-1 -antitrypsin	(iv)	Used to treat emphysema

(1) A-iii, B-i, C-iv, D-ii

(2) A-iii, B-i, C-ii, D-iv

(3) A-i, B-ii, C-iii, D-iv

(4) A-iii, B-ii, C-iv, D-i

158. At present __A__ recombinants therapeutics being marketed in India whereas __B__ recombinants therapeutics have been approved world over for human-use. **Correct** option for A and B is

	A	B
(1)	30	12
(2)	12	30
(3)	36	24
(4)	40	30

159. Identify the statements as true (T) or false (F) and select the **correct** option.

(i) Fishery is an flourishing industry meeting the ever increasing demand for only fish and fish products

(ii) In interspecific hybridisation methods, male and female animals of two different related species are mated

(iii) Mackerel and pomfrets are example of the edible fresh water fishes

(iv) Breeding between animals of the same breed is called inbreeding whereas breeding between different breeds of same species is called out-breeding

	I	II	III	IV
1.	F	T	F	F
2.	F	T	F	T
3.	F	T	T	T
4.	T	F	F	T

160. Choose the **mismatch** w.r.t. drugs and their effects/uses.

(1) Opioids – Effective sedative and pain killer

(2) Barbiturates – Help the patients cope with depression and insomnia

(3) Cocaine – Produces a sense of euphoria and increased energy

(4) Cannabinoids – Has a potent stimulating action on central nervous system

161. Which of the following statement is **incorrect**?

(1) Vaccines produced using Recombinant DNA technology, provide passive immunity to infant

(2) Colostrum has abundant antibodies (IgA) and provide passive immunity to infant

(3) Mucosa-associated lymphoid tissue constitutes about 50 per cent of the lymphoid tissue in human body

(4) For HIV infected persons, time-lag between the infection and appearance of AIDS symptoms may vary from a few months to many years

162. The uses of vaccines and immunisation programmes have enabled us to completely eradicate deadly disease like

(1) Dengue (2) Polio

(3) Diptheria (4) Small pox

163. Identify the stage of *plasmodium* which forms gametes, and taken up by female *anopheles* mosquito when bites an infected person

(1) Sporozoite

(2) Gametocyte

(3) Merozoite

(4) Gamete

164. Which of the following statement is **incorrect** for contraceptives?

(1) Barrier methods prevent conception by blocking the entry of sperms through the cervix

(2) Mode of action of implants is similar to that of pills but their effective periods are much lesser compare to later

(3) Administration of progestogens or progestogen estrogen combination within 72 hours of coitus considered as emergency contraceptives

(4) LNG-20 prevents implantation as well as alter the quality of cervical mucus to retard entry of sperms

165. Identify the **correct** set of STDs which are completely curable if detected early and treated properly.

(1) Genital herpes, HIV

(2) Genital warts, Trichomoniasis

(3) Hepatitis-B, HIV

(4) Chlamydiasis, Genital herpes

166. Which of the following ART does not involve *in vitro* fertilisation?

(1) ICSI

(2) ZIFT

(3) GIFT

(4) IUT

167. How many primary spermatocytes are required to form 1000 sperms?

(1) 1000 (2) 500

(3) 250 (4) 750

168. Select the **correct** statement w.r.t. oogenesis/menstrual cycle

(1) First polar body is formed during luteal phase of menstrual cycle

(2) Fluid filled cavity called antrum is absent in secondary and tertiary follicle

(3) The ovulatory phase is followed by the proliferative phase during which reptured graafian follicle transforms as the corpus luteum

(4) Primary oocyte within the tertiary follicle grows in size and completes its first meiotic division

169. If a male ejaculates about 300 million sperms during a coitus of which, how many of them must show normal shape, size and vigorous motility for normal fertility?

(1) 180 million

(2) 72 million

(3) 150 million

(4) 300 million

170. Read the following statements and select the **correct** option

Statement A: The process of childbirth is called parturition which is induced by complex neuroendocrine mechanism involving cortisol, estrogen and oxytocin

Statement B: By the end of the 8 weeks of pregnancy, the foetus develops limbs and digits

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is incorrect

(4) Only statement B is incorrect

171. Choose the **correct** match

(1) Insulin – Gluconeogenesis

(2) Catecholamines – Glycogenolysis

(3) Glucagon – Glycogenesis

(4) Androgens – Catabolic effects on protein and carbohydrate metabolism

172. Which of the following gastro-intestinal hormone acts on liver as well as exocrine pancreas?

(1) Cholecystokinin

(2) Gastrin

(3) Secretin

(4) Gastric inhibitory peptide

173. All of the followings interact with intracellular receptors **except**

(1) Estrogen

(2) Erythropoietin

(3) Cortisol

(4) Aldosterone

174. Tectorial membrane is structural constituent of

(1) Cirsta ampullaris

(2) Macula

(3) Organ of corti

(4) Vestibular apparatus

175. At the base of the cochlea, the ___A___ ends at the ___B___ window. Choose the option that **correctly** fill the blanks A and B

(1) Scala vestibuli, round

(2) Scala media, oval

(3) Scala vestibuli, oval

(4) Scala media, round

176. Which of the following part of the forebrain is involved in the regulation of sexual behaviour, along with hypothalamus?

(1) Cerebellum

(2) Limbic system

(3) Corpora quadrigemina

(4) Cerebral aqueduct

177. Dendrites are generally **not** present in

(1) Myelinated neurons

(2) Non-myelinated neurons

(3) Bipolar neurons

(4) White matter of cerebrum

178. Select the **odd** one out w.r.t. synovial joints

(1) Pivot joint

(2) Joint between the adjacent vertebrae

(3) Joint between the carpals

(4) Joint between the humerus and pectoral girdle

179. Which of the following secondary metabolites is a pigment?

(1) Concanavalin A

(2) Abrin

(3) Carotenoids

(4) Codeine

180. Gross calorific values of carbohydrates, fats and proteins are respectively (k Cal/g)

(1) 4.1, 5.65, 9.45

(2) 4.1, 9.45, 5.65

(3) 4.0, 4.0, 9.0

(4) 9, 4.0, 4.00

Aakash Medical | IIT-JEE | Foundations VIP HAWELI

Regd. Office :Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

NCERT BOOSTER TEST SERIES

Test – 10 (Answer Key)

1. (3)	37. (4)	73. (4)	109. (4)	145. (4)
2. (3)	38. (1)	74. (2)	110. (1)	146. (3)
3. (1)	39. (1)	75. (4)	111. (3)	147. (4)
4. (4)	40. (2)	76. (1)	112. (2)	148. (2)
5. (2)	41. (1)	77. (1)	113. (1)	149. (2)
6. (1)	42. (2)	78. (2)	114. (1)	150. (4)
7. (1)	43. (4)	79. (4)	115. (2)	151. (3)
8. (3)	44. (3)	80. (3)	116. (4)	152. (1)
9. (4)	45. (1)	81. (1)	117. (3)	153. (3)
10. (3)	46. (1)	82. (3)	118. (1)	154. (3)
11. (1)	47. (2)	83. (1)	119. (2)	155. (3)
12. (2)	48. (3)	84. (2)	120. (3)	156. (1)
13. (3)	49. (3)	85. (1)	121. (2)	157. (2)
14. (3)	50. (4)	86. (3)	122. (3)	158. (2)
15. (3)	51. (3)	87. (3)	123. (4)	159. (2)
16. (4)	52. (3)	88. (1)	124. (2)	160. (4)
17. (1)	53. (1)	89. (2)	125. (1)	161. (1)
18. (2)	54. (3)	90. (2)	126. (1)	162. (4)
19. (4)	55. (2)	91. (1)	127. (1)	163. (2)
20. (3)	56. (3)	92. (3)	128. (1)	164. (2)
21. (1)	57. (2)	93. (2)	129. (3)	165. (2)
22. (3)	58. (1)	94. (1)	130. (4)	166. (3)
23. (3)	59. (3)	95. (3)	131. (1)	167. (3)
24. (1)	60. (1)	96. (4)	132. (1)	168. (4)
25. (3)	61. (4)	97. (1)	133. (1)	169. (2)
26. (4)	62. (1)	98. (2)	134. (3)	170. (1)
27. (4)	63. (2)	99. (4)	135. (2)	171. (2)
28. (1)	64. (4)	100. (4)	136. (3)	172. (3)
29. (4)	65. (2)	101. (1)	137. (3)	173. (2)
30. (2)	66. (4)	102. (4)	138. (4)	174. (3)
31. (3)	67. (3)	103. (2)	139. (4)	175. (3)
32. (4)	68. (4)	104. (3)	140. (3)	176. (2)
33. (4)	69. (2)	105. (3)	141. (3)	177. (4)
34. (3)	70. (4)	106. (1)	142. (3)	178. (2)
35. (1)	71. (3)	107. (1)	143. (3)	179. (3)
36. (4)	72. (3)	108. (3)	144. (2)	180. (2)